A Laboratory Manual for

Database Management System (BE03000091)

B.E. Semester 3

(Computer Engineering/ Artificial Intelligence & Data science)





Directorate of Technical Education, Gandhinagar, Gujarat

Government Engineering College –Rajkot

Certificate

This is to certify that Mr./Ms. _____ _____ Enrollment No. _____ of B.E. Semester _____ Computer Engineering/Artificial Intelligence and Data Science of this Institute (GTU Code:020) has satisfactorily completed the Practical work for the subject **Database Management System (BE03000091)** for the academic year 20__-__.

Place: _____
Date: _____

Name and Sign of Faculty member

Head of the Department

Preface

Main motto of any laboratory/practical/field work is for enhancing required skills as well as creating ability amongst students to solve real time problem by developing relevant competencies in psychomotor domain. By keeping in view, GTU has designed competency focused outcome-based curriculum for engineering degree programs where sufficient weightage is given to practical work. It shows importance of enhancement of skills amongst the students and it pays attention to utilize every second of time allotted for practical amongst students, instructors and faculty members to achieve relevant outcomes by performing the experiments rather than having merely study type experiments. It is must for effective implementation of competency focused outcome-based curriculum that every practical is keenly designed to serve as a tool to develop and enhance relevant competency required by the various industry among every student. These psychomotor skills are very difficult to develop through traditional chalk and board content delivery method in the classroom. Accordingly, this lab manual is designed to focus on the industry defined relevant outcomes, rather than old practice of conducting practical to prove concept and theory.

By using this lab manual students can go through the relevant theory and procedure in advance before the actual performance which creates an interest and students can have basic idea prior to performance. This in turn enhances pre-determined outcomes amongst students. Each experiment in this manual begins with competency, industry relevant skills, course outcomes as well as practical outcomes (objectives). The students will also achieve safety and necessary precautions to be taken while performing practical.

This manual also provides guidelines to faculty members to facilitate student centric lab activities through each experiment by arranging and managing necessary resources in order that the students follow the procedures with required safety and necessary precautions to achieve the outcomes. It also gives an idea that how students will be assessed by providing rubrics.

Database Management System is a core subject of Computer Science & Engineering-Data Science course. Database is an integral part of real life application system. The course will enable student understand the different issues involved in the design and implementation of a database system. Student will learn the physical and logical database designs, database modeling, relational, hierarchical, and network models. Student will learn to use data manipulation language to query, update, and manage a database. Student will understand essential DBMS concepts such as: database security, integrity, concurrency, storage strategies etc. The students will get the hands on practice of using SQL and PL/SQL concepts.

Utmost care has been taken while preparing this lab manual however always there is chances of improvement. Therefore, we welcome constructive suggestions for improvement and removal of errors if any.

DTE's Vision

- To provide globally competitive technical education
- Remove geographical imbalances and inconsistencies
- Develop student friendly resources with a special focus on girls' education and support to weaker sections
- Develop programs relevant to industry and create a vibrant pool of technical professionals

Vision:

To bring out technically competent and socially responsible engineers.

Mission:

- To improve pedagogical skills and subject knowledge of the faculty members.
- To take cognizance of social, ethical and environmental issues.
- To nurture innovative thinking and experimentation for addressing real life problems.
- To fortify industry interaction for up gradation of the skills of students for meeting upcoming professional challenges

Vision:

To provide value-based technology education in Computer Engineering.

Mission:

- To bring out graduates who can solve challenges of industry and society by applying computing techniques.
- To develop partnership with industries, government agencies and Research and Development organizations for knowledge and resource sharing.
- To encourage faculties and students to participate in reputed conferences, workshops, seminars and other such technical activities.
- To motivate students/graduates to be entrepreneurs.
- To impart human and ethical values among students in the service of society.

Programme Outcomes (POs)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Outcomes (PEO) :

Graduates will…

PEO1: Solve real life problems related to Computing.

PEO2: Develop computing solutions to the business and societal needs.

PEO3: Pursue higher studies, carry out R&D, imbibe high degree of professionalism or become entrepreneur.

PEO4: Embrace life-long learning.

PEO5:Work and excel in a highly competitive and multicultural environment by abiding legal and ethical responsibilities.

PROGRAM SPECIFIC OUTCOMES (PSO)

Students will be able to…

PSO1: Apply the fundamental knowledge of computer engineering to develop computer programs / computer-based systems in the areas of algorithms, networking, web design, cloud computing, IoT and data analytics of varying complexity.

PSO2: Apply the core concepts of information assurance and security, system architecture, system administration, maintenance and integration, to handle real-time software projects.

PS3:Opt for start-ups by practicing the technical skill and ethical values cultivated.

Practical – Course Outcome matrix

Course Outcomes (COs):

CO_03000091.1 Understand DBMS fundamentals, database architecture, relational models, and user roles.

CO_03000091.2 Design databases using E-R diagrams, normalization, and relational schemas.

CO_03000091.3 Apply SQL and PL/SQL for database queries, constraints, procedures, and triggers.

CO_03000091.4 Use transaction management, concurrency control, and recovery techniques.

CO_03000091.5 Demonstrate query optimization, indexing, hashing, and database security

Sr. No.	Objective(s) of Experiment	CO 1	CO 2	CO 3	CO 4	CO 5
1.	Create Tables, insert Into Table, Perform Select Queries, where clause, Order by clause, In operator, Between Operator, Is Null, Is Not Null Operator and Like Operator			√		
2.	Implement Data Constraints: Add primary key, foreign key, unique key, and check constraints, define and remove integrity constraints using the ALTER TABLE command.			√		
3.	Displaying data from Multiple Tables using join operations.			√		
4.	Perform Functions & Advanced Operators.			√		
5.	Solve queries using the concept of sub query.			√		
6.	Study & Implementation of PL/SQL using Cursor.			√		
7.	Study & Implementation of PL/SQL using Trigger.			√		
8.	Study & Implementation of PL/SQL using Procedure and Function.			√		

Industry Relevant Skills

The following industry relevant competency is expected to be developed in the student by undertaking the practical work of this laboratory.

1. Will be able to design and develop a database for any application
2. Access and modify the created database as per requirement
3. Optimize data access also understand the internal function of various component of DBMS

Guidelines for Faculty members

1. Teacher should provide the guideline with demonstration of practical to the students with all features.
2. Teacher shall explain basic concepts/theory related to the experiment to the students before starting of each practical
3. Involve all the students in performance of each experiment.
4. Teacher is expected to share the skills and competencies to be developed in the students and ensure that the respective skills and competencies are developed in the students after the completion of the experimentation.
5. Teachers should give opportunity to students for hands-on experience after the demonstration.
6. Teacher may provide additional knowledge and skills to the students even though not covered in the manual but are expected from the students by concerned industry.
7. Give practical assignment and assess the performance of students based on task assigned to check whether it is as per the instructions or not.
8. Teacher is expected to refer complete curriculum of the course and follow the guidelines for implementation.

Instructions for Students

1. Students are expected to carefully listen to all the theory classes delivered by the faculty members and understand the COs, content of the course, teaching and examination scheme, skill set to be developed etc.
2. Student should develop a habit of submitting the experimentation work as per the schedule and s/he should be well prepared for the same.

Common Safety Instructions

Students are expected to

1) switch on the PC carefully (not to use wet hands)
2) shutdown the PC properly at the end of your Lab
3) carefully Handle the peripherals (Mouse, Keyboard, Network cable etc)
4) Use Laptop in lab after getting permission from Teacher

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(Progressive Assessment Sheet)

Sr. No.	Objective(s) of Experiment	Page No.	Date of performance	Date of submission	Assessment Marks	Sign. of Teacher with date
1.	Create Tables, insert Into Table, Perform Select Queries, where clause, Order by clause, In operator, Between Operator, Is Null, Is Not Null Operator and Like Operator					
2.	Implement Data Constraints: Add primary key, foreign key, unique key, and check constraints, define and remove integrity constraints using the ALTER TABLE command.					
3.	Displaying data from Multiple Tables using join operations.					
4.	Perform Functions & Advanced Operators.					
5.	Solve queries using the concept of sub query.					
6.	Study & Implementation of PL/SQL using Cursor.					
7.	Study & Implementation of PL/SQL using Trigger.					
8.	Study & Implementation of PL/SQL using Procedure and Function.					
Total						

Experiment No: 1

Experiment-1: Create Tables, insert Into Table, Perform Select Queries, where clause, Order by clause, In operator, Between Operator, Is Null, Is Not Null Operator and Like Operator

Date:
Relevant CO: CO3
Objectives: Use of where clause, Order by Clause and various operators like IN, Between,IS NULL,IS NOT NULL and Like
Equipment/Instruments: Personal Computer,MySQL

Explanation:

Create Database, Create Tables, Insert Data

Departments Table:

department_id	department_name
1	Computer Engineering
2	Electronics
3	Mechanical
4	Civil
5	IT
6	Chemical
7	Instrumentation
8	Automobile
9	Textile
10	Environmental
11	Mechatronics
12	Aerospace
13	Mining
14	Biomedical
15	Petroleum
16	Robotics
17	Agricultural
18	Architecture
19	Marine
20	Nanotechnology

Students Table:

student_id	name	dob	email	department_id
101	Khushi Shah	2004-05-10	alice@example.com	1
102	Bhavesh Patel	2003-11-23	bhavesh@example.com	1
103	Chirag Mehta	2004-01-15	chirag@example.com	2
104	Dhruvi Joshi	2003-06-08	dhruvi@example.com	3
105	Esha Trivedi	2004-09-14	esha@example.com	1
106	Farhan Shaikh	2003-12-20	farhan@example.com	2
107	Gaurav Rana	2004-03-12	gaurav@example.com	3
108	Heena Patel	2004-08-30	heena@example.com	1
109	Isha Desai	2004-07-01	isha@example.com	2
110	Jay Solanki	2003-04-25	jay@example.com	3
111	Kajal Vyas	2004-11-05	kajal@example.com	1
112	Laksh Soni	2003-10-18	laksh@example.com	1
113	Manav Panchal	2004-01-22	manav@example.com	2
114	Neha Gohil	2003-03-19	neha@example.com	2
115	Om Thakkar	2004-02-28	om@example.com	3
116	Pooja Zala	2004-07-17	pooja@example.com	1
117	Ravi Chauhan	2003-12-12	ravi@example.com	2
118	Sneha Rawal	2004-06-16	sneha@example.com	3
119	Tanvi Shah	2004-10-02	tanvi@example.com	1
120	Utsav Dave	2003-09-09	utsav@example.com	3
121	NULL	2000-12-01	NULL	2

Courses Table:

course_id	course_name	credits
201	Database Systems	4
202	Operating Systems	3
203	Microprocessors	3
204	Computer Networks	4
205	Data Structures	4
206	Software Engineering	3
207	Web Development	3
208	Machine Learning	4
209	Compiler Design	4
210	Cloud Computing	3
211	Cyber Security	3
212	IoT	3
213	AI Fundamentals	4
214	DBMS Lab	2
215	Operating Systems Lab	2
216	DS Lab	2
217	Project Phase 1	2
218	Project Phase 2	2
219	Python Programming	4
220	C Programming	4
221	DBMS Lab	2
222	NULL	3
223	DBMS Lab	3
224	IOT Lab	NULL

Faculty Table:

faculty_id	name	email	department_id
301	Prof. Kiran	kiran@college.edu	1
302	Prof. Mehul	mehul@college.edu	2
303	Prof. Rina	rina@college.edu	1
304	Prof. Tejas	tejas@college.edu	1
305	Prof. Dhara	dhara@college.edu	3
306	Prof. Nirav	nirav@college.edu	1
307	Prof. Mitali	mitali@college.edu	2
308	Prof. Arjun	arjun@college.edu	3
309	Prof. Bharti	bharti@college.edu	2
310	Prof. Harsh	harsh@college.edu	1
311	Prof. Zarna	zarna@college.edu	3
312	Prof. Kamal	kamal@college.edu	1
313	Prof. Harita	harita@college.edu	2
314	Prof. Suresh	suresh@college.edu	3
315	Prof. Reema	reema@college.edu	1
316	Prof. Ankit	ankit@college.edu	2
317	Prof. Seema	seema@college.edu	1
318	Prof. Viraj	viraj@college.edu	3
319	Prof. Hetal	hetal@college.edu	1
320	Prof. Shweta	shweta@college.edu	1

course_faculty Table:

course_id	faculty_id	semester
201	301	4
202	301	4
203	302	4
204	303	5
205	304	3
206	305	3
207	306	4
208	307	6
209	308	5
210	309	6
211	310	5
212	311	5
213	312	6
214	313	4
215	314	4
216	315	3
217	316	7
218	317	8
219	318	3
220	319	2

Enrollments Table:

enrollment_id	student_id	course_id	semester	grade
1	101	201	4	A
2	101	202	4	B
3	102	201	4	A
4	103	203	4	C
5	104	202	4	B
6	105	201	4	A
7	106	205	3	B
8	107	204	5	B
9	108	204	5	A
10	109	205	3	C
11	110	206	3	B
12	111	207	4	A
13	112	208	6	B
14	113	209	5	A
15	114	210	6	C
16	115	211	5	B
17	116	212	5	B
18	117	213	6	A
19	118	214	4	A
20	119	215	4	B

SELECT and WHERE Clauses

1. Display all student details from the students table.
2. Show the names and emails of students from the ‘Computer Engineering’ department.
3. List all courses that have more than 3 credits.
4. Find faculty members who belong to department ID 1 or 2.
5. Display all students who were born after the year 2003.

ORDER BY Clause

6. List all students sorted by their name in ascending order.
7. Show the top 5 most recent students based on date of birth.
8. List all courses ordered by credits in descending order.
9. Display faculty details sorted alphabetically by name.

IN Operator

10. List students from department IDs 1, 2, or 3.
11. Show course names for course IDs 201, 204, 207.
12. Find faculty assigned to courses 201 or 203.

BETWEEN Operator

13. Find students born between 2003 and 2004.
14. List all courses that have credits between 3 and 4.
15. Display enrollments for semesters between 4 and 6.

IS NULL / IS NOT NULL Operators

16. Find any students with missing email addresses (if any).
17. List all faculty who have email addresses filled.
18. Show enrollments where grade is not yet assigned (assume NULL grades).

LIKE Operator

19. Find students whose names start with 'A'.
20. Find students whose names end with 'i'.
21. List all faculty whose names contain 'a' anywhere.
22. Find courses that have the word ‘Data’ in the course name.
23. Show departments whose name starts with 'C'.

Combined Conditions

24. Display students in Computer Engineering born between 2003 and 2004.
25. List course names and faculty names for courses having more than 3 credits.

Safety and necessary Precautions:

1. Make sure the database server is started
2. Login with correct user ID and password in MySQL.

Procedure:

1. Get MySQL prompt
2. Login with correct userID and password

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment No: 2

Implement Data Constraints: Add primary key, foreign key, unique key, and check constraints, define and remove integrity constraints using the ALTER TABLE command.
Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Detail Explanation:

Update Queries (1–5)

1. Update the email address of the student with student ID 101 to alice.shah@college.edu.
2. Update all students from department ID 3 to department ID 2.
3. Change the number of credits for the course "Database Systems" to 5.
4. Change the name of the faculty with faculty ID 302 to "Prof. Mehul Desai".
5. Update all enrollments where grade is 'C' and change it to 'B'.

Alter Table – Add, Modify, Drop Column (6-10)

6. Add a new column phone of type VARCHAR(15) to the students table.
7. Modify the phone column in the students table to restrict it to VARCHAR(10).
8. Remove the phone column from the students table.
9. Add a column gender of type CHAR(1) to the faculty table.
10. Modify the gender column in the faculty table to be VARCHAR(6) to store values like "Male" or "Female".
11. Remove the gender column from the faculty table.

Alter Table – With Defaults and Constraints (12-15)

12. Add a new column semester (INT) to the courses table.
13. Modify the semester column to set its default value to 4.
14. Remove the semester column from the courses table.
15. Modify the name column in the students table to ensure it cannot be NULL.

Create and Alter Table – Primary Key and Foreign Key (16)

16. Create a new table named test_enrollments with columns enroll_id, student_id, and course_id.
    Then, perform the following actions:
    - Add a primary key on enroll_id.
    - Add a foreign key on student_id referencing students(student_id).
    - Add a foreign key on course_id referencing courses(course_id).

Delete Queries (17-22)

17. Create a new table named test_enrollments with columns enroll_id, student_id, and course_id.

18. Delete the enrollment record with enrollment ID 20.
19. Delete all students who belong to course_ ID 9.
20. Delete all faculty members who belong to Course_ID 3.
21. Delete all students whose course id is not specified (NULL).
22. Delete all courses having student_id=1.

Truncate Table Queries

23. Remove all records from the test_enrollments table without deleting the table structure.

Drop Table Queries

24. Drop the table test_enrollments if it exists.

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

References used by the students:

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment No: 3

Displaying data from Multiple Tables using join operations.

Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Detail Explanation:

INNER JOIN (15 Queries)

1. Display student names along with their department names.
2. Display faculty names with the names of their departments.
3. Show student names and the course IDs they are enrolled in.
4. Display course names with their corresponding department IDs.
5. Show course names and the names of faculty members who teach them.
6. List faculty names and the course IDs they are assigned to.
7. Show student names and their grades for the courses they are enrolled in.
8. Display course names and the student IDs of enrolled students.
9. Show student IDs along with their department names.
10. Show faculty IDs and the names of the departments they belong to.
11. Display course names along with the department names they are part of.
12. List student names with the course names they are enrolled in.
13. Show faculty names and the names of courses they are teaching.
14. Display student names along with the grades they received.
15. List course names with faculty IDs teaching those courses.

Other JOINs (5 Queries)

16. **LEFT JOIN**: List all students with their enrolled course IDs (if any).
17. **LEFT JOIN**: Show all courses along with their assigned faculty IDs (if any).
18. **RIGHT JOIN**: Display all departments with the names of faculty in each (if any).
19. **LEFT JOIN**: Show all faculty names with the course IDs they are teaching (if any).
20. **SELF JOIN**: List pairs of students who belong to the same department.

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

References used by the students:

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment No: 4

Perform Functions & Advanced Operators.
Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Detail Explanation:

Aggregate Functions (1–6)

1) Display the total number of students in the database.
2) Show the average grade (assuming letter grades are mapped to numbers) for all enrollments.
3) Find the total number of students in each department.
4) Display the maximum number of credits assigned to any course.
5) Count how many courses are offered by each faculty member.
6) Show the minimum student ID in the students table.

Single Row Functions (7–12)

7) Display the names of all students in uppercase.
8) Show the length of each course name.
9) Display the first 3 characters of each department name.
10) Show the current date and time using MySQL function.
11) Display the email domain (portion after @) of each student.
12) Show the name of each student along with their name reversed.

GROUP BY Clause (13–17)

13) Group students by department and display the count of students in each.
14) Group courses by credit value and count how many courses fall under each credit.
15) Group faculty by department and display the number of faculty members per department.
16) Show total number of enrollments per course.
17) Display the average student ID (numerically) per department.

HAVING Clause (18–19)

18) Display departments with more than 3 students.
19) List faculty members who are assigned more than 2 courses.

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

References used by the students:

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment-5

Solve queries using the concept of sub query.

Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Sub Queries (1-10):

1) Find the names of students who belong to the department with ID 2.
2) Find the names of courses with more credits than the average course credits.
3) List the names of students who are enrolled in course ID 101.
4) Display the names of faculty who are from the department named "Computer Engineering".
5) Show the names of students who are not enrolled in any course.
6) Display course names that have been assigned to any faculty.
7) Show the name of the department with the highest number of students.
8) List the names of students who have received a grade "A".
9) Find faculty who are not assigned to any course.
10) List courses which have not been taken by any student.

EXISTS Operator (11-12)

11) Display all students who have at least one enrollment (i.e., their ID exists in the enrollments table).
12) Display all faculty members who are assigned to at least one course (using EXISTS with course_faculty table).

ANY / ALL Operators (13-16)

13) Show the students whose ID is greater than **any** student ID in department 2.
14) Show the students whose ID is greater than **all** student IDs in department 1.
15) Display courses that have fewer credits than **any** course offered in semester 6.
16) Display departments that have fewer students than **all** other departments.

Solution: (write your queries here)

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment-6

Study & Implementation of PL/SQL using Cursor.

Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL
Consider below table that already created in Practical-3 and 4:
No need to create below table again:

Detail Explanation:

Departments Table

```
CREATE TABLE departments (
   department_id INT PRIMARY KEY,
   department_name VARCHAR(100));
```

Students Table

```
CREATE TABLE students (
   student_id INT PRIMARY KEY,
   name VARCHAR(100),
   dob DATE,
   email VARCHAR(100),
   department_id INT,
   FOREIGN KEY (department_id) REFERENCES departments(department_id));
```

Courses Table

```
CREATE TABLE courses (
   course_id INT PRIMARY KEY,
   course_name VARCHAR(100),
   credits INT);
```

Faculty Table

```
CREATE TABLE faculty (
   faculty_id INT PRIMARY KEY,
   name VARCHAR(100),
   email VARCHAR(100),
   department_id INT,
   FOREIGN KEY (department_id) REFERENCES departments(department_id));
```

Create cursor:

1. Create a PL/SQL block that uses a cursor to list all students from a specific department (e.g., 'Computer Engineering').

2. To use an explicit cursor to fetch and display faculty Names.

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

References used by the students:

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment-7

Study & Implementation of PL/SQL using Trigger.
Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Detail Explanation:

Consider the table which are already created in Practical-3 and 4.

No need to create below table again:

Departments Table:

```
CREATE TABLE departments (
   department_id INT PRIMARY KEY,
   department_name VARCHAR(100)
);
```

Students Table:

```
CREATE TABLE students (
   student_id INT PRIMARY KEY,
   name VARCHAR(100),
   dob DATE,
   email VARCHAR(100),
   department_id INT,
   FOREIGN KEY (department_id) REFERENCES departments(department_id)
);
```

1) Create Trigger to log student insertion.

```
CREATE TABLE student_audit(
id NUMBER GENERATED ALWAYS AS IDENTITY,
student_id NUMBER,
action_date DATE,
action VARCHAR2(20));
```

2) **Write a trigger for student table, which calculates total of 3 subjects and percentage and stores it in table for that particular record after insert.**

```
create table student_detail(
tid number (4),
name varchar2 (20),
subj1 number(10),
subj2 number(10),
subj3 number(10),
total number(10),
per number(10));
```

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						

Experiment-8

Study & Implementation of PL/SQL using Procedure and Function.

Date:

Relevant CO: CO3

Equipment/Instruments: Personal Computer, MySQL

Detail Explanation:
Consider tables which are already created in Practical-3 and 4:
No need to create below table again:

Table structure:

Departments Table

```
CREATE TABLE departments (
   department_id INT PRIMARY KEY,
   department_name VARCHAR(100)
);
```

Students Table

```
CREATE TABLE students (
   student_id INT PRIMARY KEY,
   name VARCHAR(100),
   dob DATE,
   email VARCHAR(100),
   department_id INT,
   FOREIGN KEY (department_id) REFERENCES departments(department_id)
);
```

Courses Table

```
CREATE TABLE courses (
   course_id INT PRIMARY KEY,
   course_name VARCHAR(100),
   credits INT
);
```

Faculty Table

```
CREATE TABLE faculty (
   faculty_id INT PRIMARY KEY,
   name VARCHAR(100),
   email VARCHAR(100),
   department_id INT,
   FOREIGN KEY (department_id) REFERENCES departments(department_id)
);
```

Course_faculty Table

```
CREATE TABLE course_faculty (
   course_id INT,
   faculty_id INT,
   semester INT,
   PRIMARY KEY (course_id, faculty_id),
   FOREIGN KEY (course_id) REFERENCES courses(course_id),
   FOREIGN KEY (faculty_id) REFERENCES faculty(faculty_id));
```

Perform below PLSQL block:

1) Create a procedure that takes student_id as input and print Details.
2) Create a procedure to assign a faculty member to a course.
3) Write PLSQL Function to calculate and return student's age from date of birth.

Solution: (write your queries here)

Suggested Reference:

1. "Understanding SQL", Martin Gruber, BPB
2. "SQL- PL/SQL", Ivan bayros

References used by the students:

Rubric wise marks obtained:

Knowledge of subject (2)		Programming Skill		Team work (2)		Communication Skill (2)		Ethics(2)	
Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Satisfactory (1)	Good (2)	Satisfactory (1)	Good (2)	Average (1)

Rubrics	**1**	**2**	**3**	**4**	**5**	**Total**
Marks						